

JOHN (CHANG WON) LEE

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EDUCATION

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| University of Toronto
PhD in Aerospace Engineering with a specialization in Robotics
• Advisor: Steven L. Waslander | Toronto, Canada
Sep 2025 – Present |
| University of Toronto
MAsc. in Aerospace Engineering with a specialization in Robotics
• Advisor: Steven L. Waslander | Toronto, Canada
Sep 2023 – Sep 2025 |
| University of Toronto
BAsc. in Engineering Science (Machine Intelligence Major) | Toronto, Canada
Sep 2018 – Jun 2023 |

SELECT PUBLICATIONS & SUBMITTED WORK

- **C. W. Lee**, S. Leveugle, P. Grouchy, C. Langley, S. Stolpner, J. Kelly, and S. L. Waslander, “FlowCLAS: Enhancing Normalizing Flow-Based Anomaly Segmentation Via Contrastive Learning,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. [pdf]
- S. Leveugle, **C. W. Lee**, S. Stolpner, C. Langley, P. Grouchy, S. L. Waslander, and J. Kelly, “A Photorealistic Dataset and Vision-Based Algorithm for Anomaly Detection During Proximity Operations in Lunar Orbit,” *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 2418–2425, 2026. [pdf]
- **C. W. Lee** and S. L. Waslander, “UncertaintyTrack: Exploiting Detection and Localization Uncertainty in Multi-Object Tracking,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 4946–4953. [pdf]
- J. Deery, **C. W. Lee**, and S. L. Waslander, “ProPanDL: A Modular Architecture for Uncertainty-Aware Panoptic Segmentation,” in *2023 20th Conference on Robots and Vision (CRV)*, 2023, pp. 137–144. [pdf]

PROFESSIONAL & RESEARCH EXPERIENCE

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| Graduate Student Researcher
University of Toronto Robotics Institute | Sep 2023 – Present
Toronto, Canada |
| <ul style="list-style-type: none">• Developed FlowCLAS, a normalizing flow-based anomaly segmentation method that incorporates contrastive learning and outlier exposure, achieving state-of-the-art performance on multiple benchmarks (published at WACV 2026).• Developed an automated Blender-based pipeline to generate a large-scale photorealistic lunar-orbit dataset, establishing a benchmark for visual anomaly detection and segmentation in space robotics (published in RA-L 2026).• Designed UncertaintyTrack, a 2D multi-object tracker that leverages learned localization uncertainty to improve robustness under occlusions and non-linear motion (published at ICRA 2024). | |
| Software Engineer Intern
16 Bit | Jan 2023 – Apr 2023
Toronto, Canada |
| <ul style="list-style-type: none">• Engineered a PyTorch-based bone mineral density (BMD) test analysis tool, combining object detection and OCR, reducing analysis time for hundreds of daily cases from minutes to seconds.• Deployed BMD test analysis pipeline using ONNX optimization and Docker containerization, enabling seamless integration with 16 Bit’s existing medical imaging services. | |

Research Assistant

University of Toronto

Dec 2021 – Aug 2022

Toronto, Canada

- Implemented and benchmarked state-of-the-art object detection and segmentation models in PyTorch, establishing performance baselines for long-range traffic detection and panoptic segmentation research (published at CRV 2023).
- Developed and executed MATLAB-based camera calibration pipeline to process 100+ hours of traffic footage, enabling high-quality data annotation for autonomous driving perception systems.

Software Engineer Intern

The Citco Group Limited

May 2021 – Apr 2022

Toronto, Canada

- Deployed a Dockerized search engine for financial application logs by leveraging the Elastic stack and custom regex patterns. Enhanced log searchability by around 500% and curtailed service downtime.
- Employed character recognition libraries and a custom REST API to effectively classify and organize financial documents within a PostgreSQL database. Accelerated document lookups for enhanced operational efficiency.
- Developed automated tests in Javascript using the Cypress framework, enabling the seamless and precise functionality of Citco's investment platform.
- Collaborated effectively with cross-functional teams by participating in daily scrums and utilizing other Agile methodologies to drive successful project outcomes.

TEACHING

- **Teaching Assistant**, Robotics Capstone Design (ROB498) Jan 2026 – Apr 2026
University of Toronto, Institute for Aerospace Studies
- **Teaching Assistant**, Mathematics for Robotics (ROB310) Sep 2023 – Dec 2025
University of Toronto, Institute for Aerospace Studies
 - Taught in Fall 2023, Fall 2024, and Fall 2025.
- **Teaching Assistant**, Artificial Intelligence (ROB311) Jan 2025 – Apr 2025
University of Toronto, Institute for Aerospace Studies
- **Teaching Assistant**, Introduction to Mobile Robotics (CSC477) Jan 2024 – Apr 2024
University of Toronto, Department of Mathematical & Computational Sciences
- **Teaching Assistant**, Linear Algebra for Engineering (MAT188) Sep 2022 – Dec 2022
University of Toronto, Department of Mathematics

ACTIVITIES & SOCIETIES**aUtoronto** Software Engineer

Aug 2021 – Apr 2023

- Engineered a robust 3D multi-object tracking system for autonomous driving in C++/Python using Extended Kalman Filters and Hidden Markov Models, increasing accuracy and efficiency by 10% and 5%.
- Integrated the tracking system into our autonomous vehicle's ROS-based perception pipeline, contributing to the team's first-place victories across multiple categories in the SAE AutoDrive Challenge II.

University of Toronto Engineering Student Ambassador

Aug 2019 – Aug 2022

- Participated in prospective student engagement as an Engineering Science representative at key faculty events, including program orientations, information sessions, and club fairs.

University of Toronto Engineering Student Mentor

Sep 2020 – Aug 2021

- Guided first-year engineering students in academic success strategies and career development, facilitating connections to institutional resources and support services.

Engineering Science Education Conference (ESEC) Web Designer & Organizer Jun 2020 – Jan 2021

- Designed and developed the conference website featuring keynote profiles, event details, and preparation resources for Engineering Science Education Conference 2021. [link]
- Collaborated with the organizing committee to coordinate industry speakers who shared insights into diverse engineering fields to guide Engineering Science students in making informed specialization choices.

HONORS, AWARDS & CERTIFICATES

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| • Dean's Honor List | 2018 – 2023 |
| • Natural Sciences and Engineering Research Council of Canada (NSERC)
Undergraduate Student Research Award (USRA) — \$7500 CAD | 2020, 2022 |
| • University of Toronto Scholar's Award — \$7500 CAD | 2018 |
| • Official Bilingual Certificate (English, French) | 2018 |

LANGUAGES

English (Native), **French** (Native), **Korean** (Native)

LINKS

- **Personal Website:** <https://johncwlee.github.io/>
- **Linkedin:** <https://www.linkedin.com/in/lee-cw-john/>
- **ESEC Website:** <https://engsci.utoronto.ca/experience/esec/esec-2021-home/>